MD RAFI ISLAM

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Availability: January 2027 (Ph.D. expected December 2026)

PROFESSIONAL SUMMARY

Embedded systems engineer with five years of experience building firmware, hardware validation workflows, and sensor data pipelines for multi-sensor wearable devices. Bare-metal C across three microcontroller families (STM32 Cortex-M4, MSP430, nRF52832); owns work end to end, from schematic revision and board bring-up through firmware, wireless communication, and data validation. Built and documented a 90-device hardware characterization library and validated released open-source software to capture 99.46 percent of actual puff duration.

TECHNICAL SKILLS

Embedded Firmware: Bare-metal C, C++, interrupt-driven architecture, DMA-based data transfer, low-power design, timing-critical control, 64-bit extended POSIX timestamping, SPI flash logging, binary log format design, BLE DFU firmware update handling

Microcontrollers and Platforms: STM32 (Cortex-M4, specifically STM32L476), MSP430 (specifically MSP430G2433), nRF52832, ESP32

Real-Time Operating Systems: FreeRTOS (basic), Zephyr (familiar)

Hardware Interfaces and Protocols: SPI (master and slave), I2C, UART, ADC, GPIO, Bluetooth Low Energy (GATT, GAP, DFU)

Hardware Bring-Up, Debug, and Validation: Oscilloscope-based signal inspection, JTAG and SWD debugging, hand soldering and board rework, PCB modification, schematic revision, sensor calibration, mixed-signal board bring-up; enclosure co-design in Autodesk Fusion 360 and Afinia H800 fused-filament-fabrication 3D printing

Test, Characterization, and Validation: Multi-device characterization campaigns, electromechanical test fixtures, unattended multi-day device logging, regression validation after firmware changes, binary log decoding and verification, executable-versus-script output consistency checking

Sensing and Signal Processing: IMUs, load cells, thermistors, capacitive touch, electromagnetic sensing; filter design, feature extraction, timestamp synchronization across sensor streams

Machine Learning and Validation: 1D-CNN encoders, CNN-BiLSTM, 1D U-Net time-series segmentation, dilated residual 1D CNNs, rank-based contrastive learning (Rank-N-Contrast), YOLOv5, gradient boosting (XGBoost, LightGBM), random forest; leave-one-subject-out and nested cross-validation, data-leakage control through per-fold refitting

Programming Languages and Frameworks: C, C++, Python, MATLAB; PyTorch, TensorFlow, Keras, scikit-learn, NumPy, SciPy, pandas, matplotlib

Development Tools: Git, GitHub, STM32CubeIDE, Code Composer Studio, Segger Embedded Studio, EasyPC Schematic and PCB Design, Autodesk Fusion 360; Claude Code and Codex (daily practice), agent-assisted and manual code review

PROFESSIONAL EXPERIENCE

Graduate Research Assistant | The University of Alabama | Tuscaloosa, AL | January 2021 – Present
Department of Electrical and Computer Engineering. Embedded firmware, hardware test infrastructure, and sensor data systems across NSF- and NIH-funded wearable device programs. Taught surface-mount soldering to about 7 undergraduates on Capstone Design teams and helped them debug microcontroller code and circuit boards; trained the clinicians and clinical researchers who collected study data to use the sensors and the data-collection GUI.

Lecturer | Premier University | Chittagong, Bangladesh | March 2018 – December 2020
Department of Electrical and Electronic Engineering. Taught Digital Signal Processing and Control Systems. Supervised two undergraduate student groups' final thesis projects.

ENGINEERING PROJECTS

FRIENDS-2 STM32WB5 Sensor Board Firmware: Dual-Core Embedded Software, Storage, and USB | github.com/mdrafiislam/FRIENDS_2_STM32WB5

Funding: NIH/NCI Award No. R01CA266170 | *January 2026 – Present* | *Role: lead developer, AI-assisted with Claude Code*
Migrates the FRIENDS vaping monitor to the STM32WB5 processor so that sensor data can be streamed rather than only logged to onboard memory.

- Developed the complete bare-metal C application firmware for a custom sensor board built on the STM32WB5MMG dual-core wireless module, splitting work across the Cortex-M4 application core and the Cortex-M0+ BLE core.
- Kept interrupt latency bounded: the ISR sets flags only, deferring SD card and USB work to the main loop.
- Built SD card logging on FatFs over SPI with a custom disk I/O layer; implemented runtime USB class switching between CDC

virtual COM port and USB Mass Storage.
- Implemented BLE streaming through the STM32_WPAN stack running on CPU2, with GATT notifications of sensor samples at 20 Hz, and software-triggered USB DFU into the ST system bootloader.
- Diagnosed and fixed a dual-core clock ownership fault by claiming the USB 48 MHz clock through a hardware semaphore, preventing the BLE core from powering it down.

FRIENDS Wearable Vaping Behavior Monitor: Firmware, Hardware Bring-Up, and Data Systems

Funding: NIH/NCI Award No. R01CA266170 | 2024 – Ongoing | Role: co-developer

- Co-developed bare-metal MSP430G2433 firmware for SPI flash logging, UART protocols, and real-time electromagnetic, capacitive-touch, and NTC thermistor sensing.
- Implemented hardware-timer-based 64-bit extended POSIX timestamping at 1/32768 s (about 30.5 microsecond) resolution, with low-power modes supporting continuous multi-day (up to 10-day) free-living data collection.
- Executed schematic updates, PCB modifications, sensor calibration, and mixed-signal debugging for hardware bring-up.
- Tested and documented 70-plus ENDS devices using the VM-FRIENDS electromechanical vaping machine; built the resulting compatibility library, which now covers 90-plus devices.
- Designed and executed fault-injection testing of the data pipeline, and verified through automated file comparison that every injected fault was detected and reported.
- Validation surfaced 17 electromagnetic false positives across the 24-hour protocol, removed by a 1.0 s minimum-duration filter, and confirmed timestamp-conversion drift below 10 s over a full day. Re-runs the device validation suite after firmware changes.
- Co-developed and validated the open-source Python FRIENDS GUI (github.com/eszsonic/FRIENDSGUI); validated it to capture 99.46 percent of actual puff duration across 72 puffing events (Journal of Open Research Software, 2026).

BLE Firmware and Low-Power Optimization for Wearable Sensors | github.com/mdrafiislam/BLE_nRF52832

Funding: NSF Grant No. 1734501 | 2023 – 2024 | Role: firmware developer

- Programmed the nRF52832 for high-throughput BLE communication at 8192 bit/s, optimizing GATT and GAP performance and DFU handling.
- Reduced current consumption from 1.6 mA to 180 microamps during transmission, and from 50 microamps to 6 microamps in advertising mode, extending device runtime for continuous free-living data collection.
- Encountered and worked within the RAM limits of the nRF52832 during BLE firmware development.

Smart Lacelock Wearable Platform: Embedded Systems, Energy Expenditure, and Mobility Assessment

Funding: NSF Grant No. 1734501 | 2021 – Ongoing | Role: co-designer, then project lead

- Co-designed the STM32L476-based embedded hardware and firmware stack for a shoe-mounted load-cell plus 9-DOF IMU wearable (magnetometer not recorded) operating at 512 Hz, implementing interrupt- and DMA-driven acquisition, synchronized timestamping, and real-time BLE streaming.
- Leads a metabolic-equivalent (MET) estimation study in 12 adults, referenced breath-by-breath against Cosmed K5 indirect calorimetry over a 14-activity protocol; the reference convolutional network reached a per-participant mean absolute error of 0.614 MET (Lin's concordance correlation coefficient 0.904) in unseen participants, with every fitted stage refit inside each leave-one-subject-out fold on training participants only.
- First-authored a deep-learning pipeline for sit-to-stand detection in 16 community-dwelling older adults; reached a pooled event F1 of 0.787 under leave-one-participant-out validation. Code: github.com/mdrafiislam/smart_lacelock_sist.

Vision-Based Human Motion Intent Recognition for Assistive Devices

Funding: NSF Grant No. 1734501 | 2022 – 2023 | Role: first author and lead developer

- First-authored a YOLOv5 plus gradient-boosting intent-recognition system that identifies stair engagement versus avoidance from egocentric video, achieving 97.69 percent recognition at least two steps before the walking-to-stair-climbing mode transition (Sensors, 2023).

EDUCATION

Ph.D., Electrical Engineering | The University of Alabama, Tuscaloosa, AL | GPA 3.92 | expected December 2026

MS, Electrical Engineering | The University of Alabama, Tuscaloosa, AL | GPA 3.92 | May 2026

BSc, Electrical and Electronic Engineering | Chittagong University of Engineering and Technology, Chittagong, Bangladesh | GPA 3.63 | November 2017

PUBLICATIONS, REVIEW, AND AWARDS

5 peer-reviewed journal papers, 5 peer-reviewed conference papers, and 1 book chapter; separately, 1 journal manuscript under review and 1 preprint. First-author journal papers in IEEE Sensors Journal (2025) and Sensors (2023). Full list on Google Scholar.

Reviewer for IEEE journals and conferences (6 manuscripts). Poster competition, 2nd place, The University of Alabama, 2023 and 2024.